

Canadian Math Olympiad 1971 — P10

Jonathan Kasongo

April 10, 2026

Canadian Math Olympiad 1971 — P10

Suppose that n people each know exactly one piece of information, and all n pieces are different. Every time person A phones person B , A tells B everything that A knows, while B tells A nothing. What is the minimum number of phone calls between pairs of people needed for everyone to know everything? Prove your answer is a minimum.

Solution

The answer is $2(n - 1)$. Finding a construction is easy. There are two main steps.

Firstly we show that it takes at least $n - 1$ phone calls to have 1 person know all bits of information. For someone to know all pieces of information, each of the other $n - 1$ people must've phoned someone else at least one time, otherwise a piece of information would still be known uniquely by exactly one person.

Now once we have someone that knows all pieces of information, for the remaining $n - 1$ people to know everything it takes at least one phone call for them to know everything. Thus in each case we add up to get at least $2(n - 1)$ phone calls. ■